Activity #2

Project Prerequisites & System Design

PantriHub - Shared Inventory Platform for Food Banks

ENSE 281 - Software Engineering Management | Group B | March 4, 2026

Today's Agenda

- Problem Recap & Stakeholders
- Stakeholder Engagement Plan
- Team Roles & RACI Matrix
- Project Scope & Requirements
- MVP Strategy & Kanban Board
- System Architecture (MVC + DDD)
- Lo-Fi & Hi-Fi Prototypes
- User Questionnaire & Feedback
- Self-Reflection

10-15 min report-out

Stakeholder Register

Stakeholder	Role	Power	Interest	Support
Project Team	Developers	High	High	**Supportive**
Food Bank Mgmt	System Owner	High	High	**Supportive**
Food Bank Staff	System Users	Medium	High	**Supportive**
Food Bank Clients	Beneficiaries	Low	High	**Supportive**
Dr. Tim Maciag	Sponsor	High	High	Neutral

Key insight: Food bank staff (daily users) confirmed the coordination problem through direct interviews in Regina, Calgary, and Vancouver.

Stakeholder Engagement Plan

Project Team

Pay Close Attention

Regular meetings, clear objectives, deadlines tracked

Food Bank Mgmt

Pay Close Attention

Progress updates aligned with project goals

Food Bank Staff

Meet Requirements

Feedback-driven design, accessible interface

Dr. Tim Maciag

Meet Requirements

On-time deliverables, incorporate feedback

Team Roles & RACI

Deliverable	Shivam	Ethan	Bushra	Thomas	Ishan
Roles & Responsibilities	**R/A**	C	C	C	C
Requirements Doc	**R/A**	**R**	C	C	C
Scope Document	C	**R/A**	C	C	C
Stakeholder Register	C	C	C	**R/A**	C
Stakeholder Engagement	C	C	C	**R/A**	C
RACI Matrix	C	C	C	C	**R/A**
Meeting Agenda / Minutes	C	C	**R/A**	I	I
UML Diagrams	**R**	**R/A**	**R**	C	C
Lo-Fi / Hi-Fi Prototypes	**R**	**R**	**R**	**R**	**R**
Presentation	**R/A**	**R**	**R**	**R**	**R**

R = Responsible A = Accountable C = Consulted I = Informed

Project Scope

In Scope

- CRUD operations for inventory items
- User authentication & role access
- Shared inventory dashboard
- Search & filter by name, category, location, expiry
- Database integration (MongoDB)
- Source code, README, full documentation
- Scalable to Saskatchewan food banks

Out of Scope

- National / international food bank coverage
- Enterprise-level security (simple role access)
- Mobile app or other platform support
- Real-time notification system
- Payment or donation processing

Key Requirements

Authentication (FR-01 to FR-03)

- Register with org name, email, password, location
- Secure login/logout
- Password reset via email

Inventory (FR-04 to FR-07)

- Add items: name, category, qty, unit, expiry
- Update & delete items
- Mark available/unavailable

Dashboard (FR-08 to FR-10)

- Shared view across all locations
- Show location, qty, category, expiry
- Highlight items expiring within 7 days

Search & Filter (FR-11 to FR-14)

- Search by item name
- Filter by category, location, expiry range

MVP Strategy

Our MVP focuses on the core problem: making food bank inventory visible across locations.

MVP 1

Core CRUD + Auth

User registration, login, add/edit/delete inventory items

MVP 2

Shared Dashboard

Cross-location inventory view with expiry highlighting

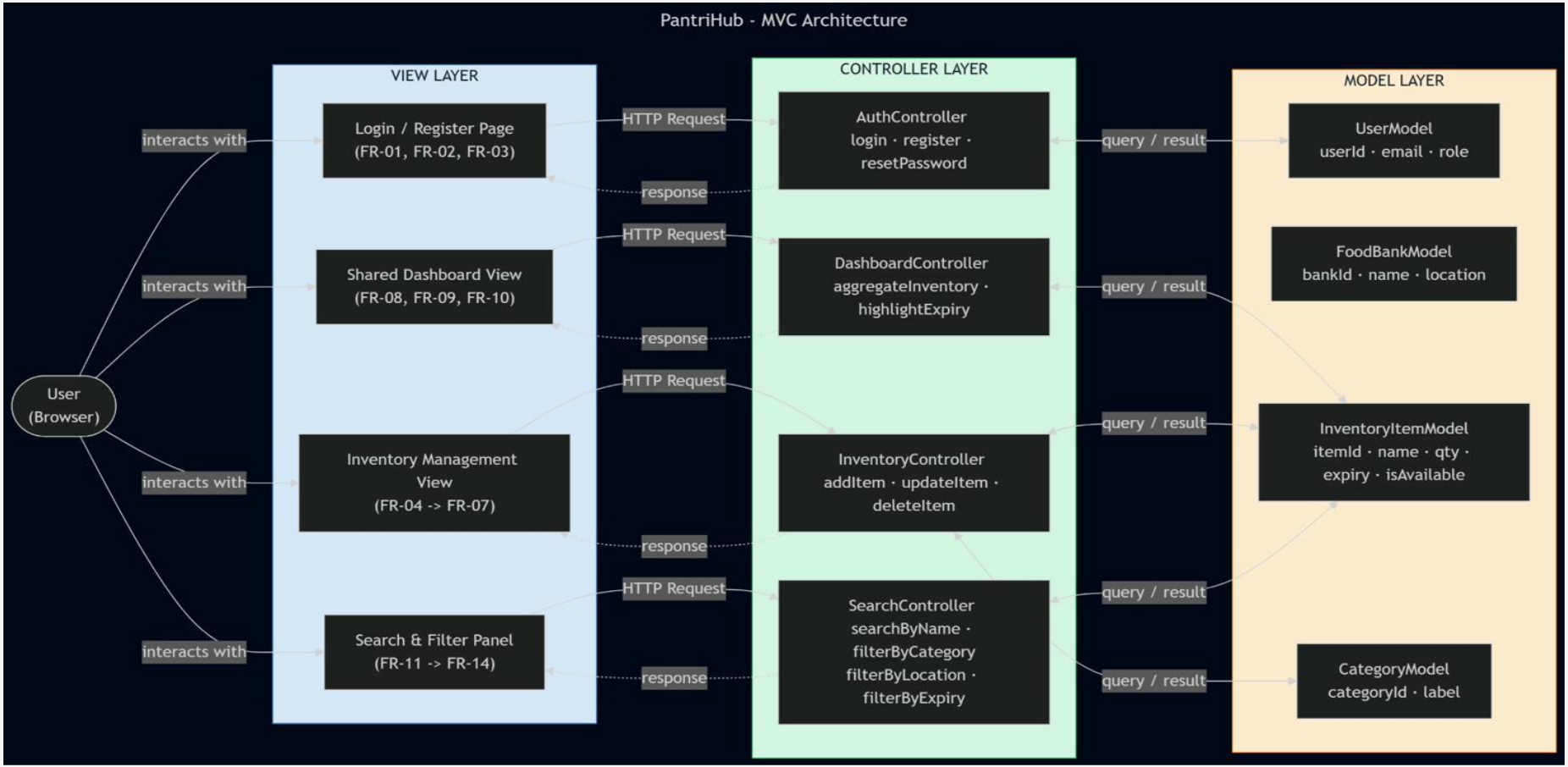
MVP 3

Search & Filter

Find items by name, category, location, expiry date range

All MVPs tracked on GitHub Project Board (Kanban) with user stories mapped to requirements.

System Architecture: MVC + Domain-Driven Design



Domain Models

UserModel

```
userId, email,
password,
role, orgName
```

FoodBankModel

```
bankId, name,
location, contactInfo
```

InventoryItemModel

```
itemId, name, qty,
unit,
category, expiry,
isAvailable,
bankId (FK)
```

CategoryModel

```
categoryId, label
(e.g. canned, dairy,
produce)
```

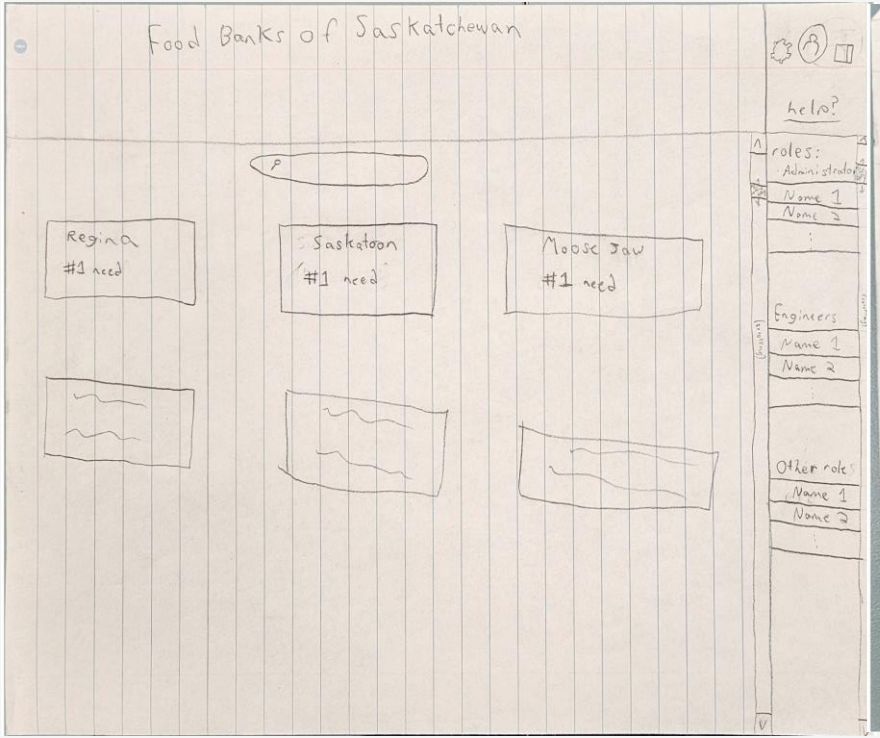
Key Relationships

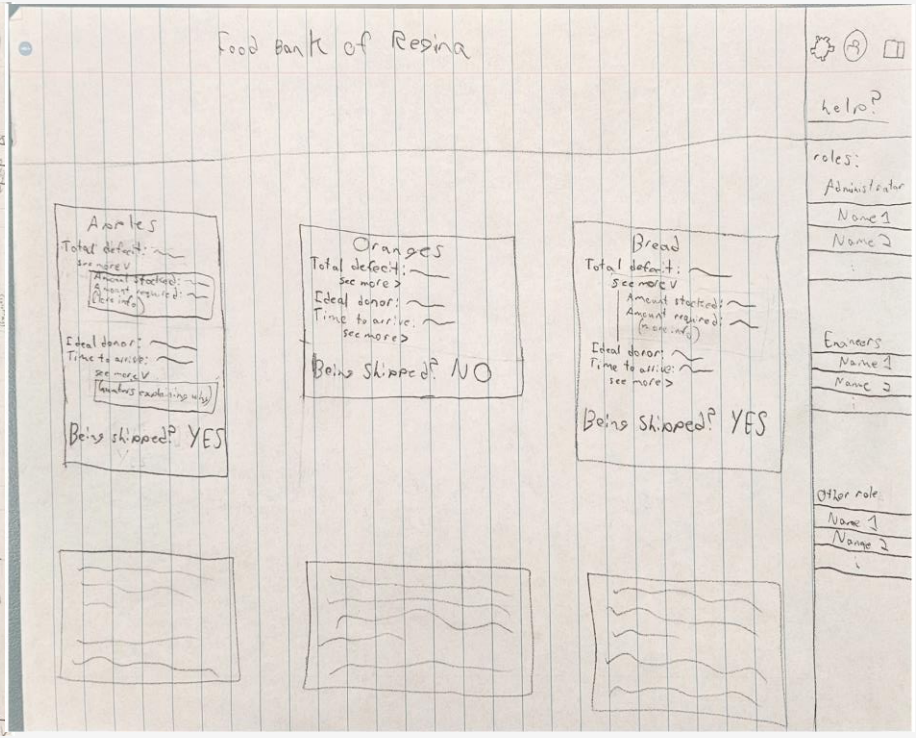
- A User belongs to one FoodBank
- A FoodBank has many InventoryItems
- Each InventoryItem has one Category
- The Shared Dashboard aggregates across all FoodBanks

Domain-Driven Design: Each model maps to a real-world concept in the food bank domain.

Lo-Fi Prototypes

Each team member sketched individual prototypes, then converged on key screens.





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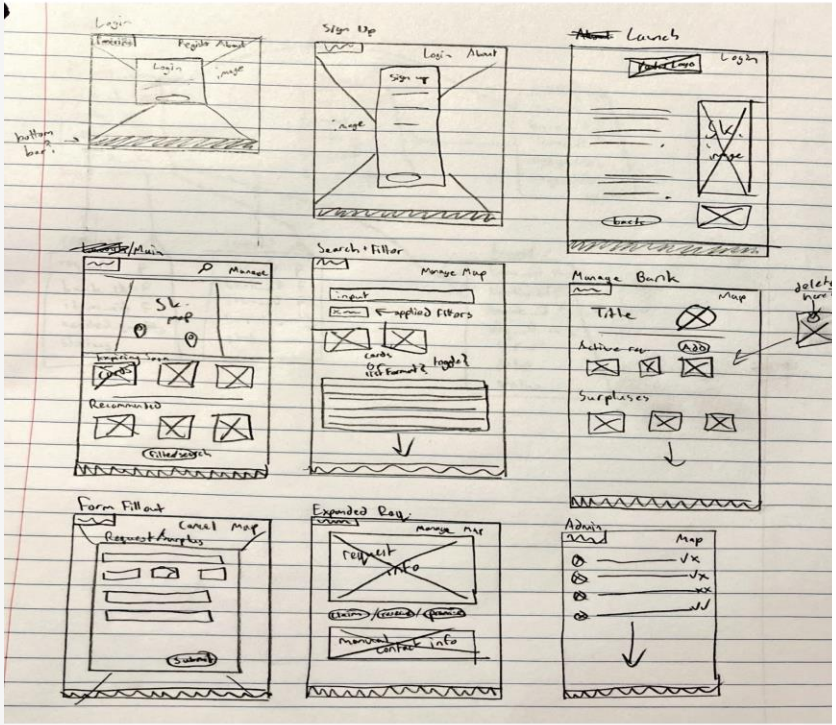
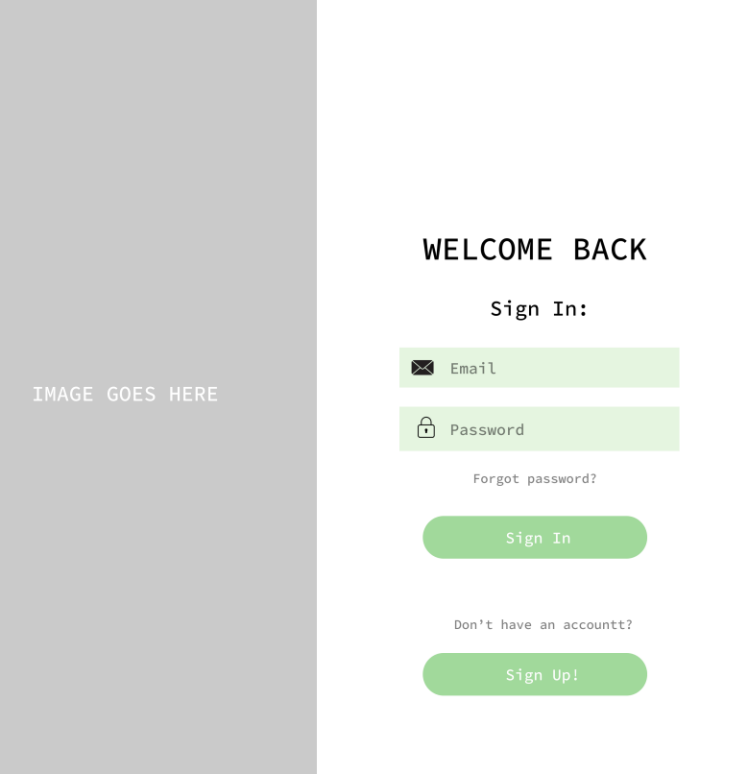
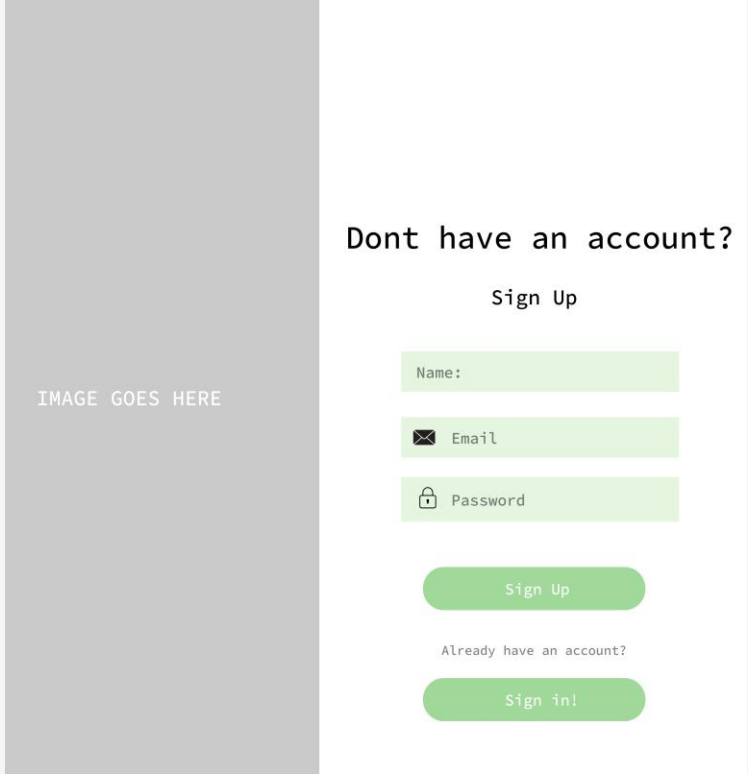


Hi-Fi Designs

Refined designs based on team feedback and usability principles.





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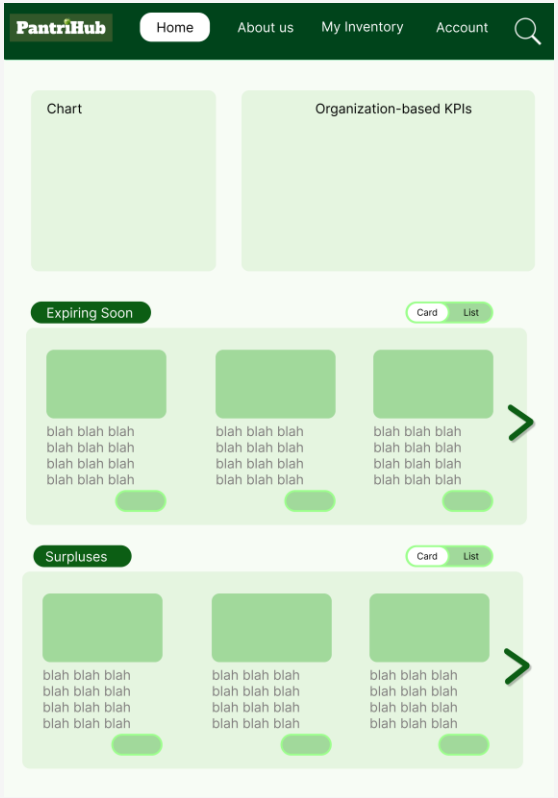

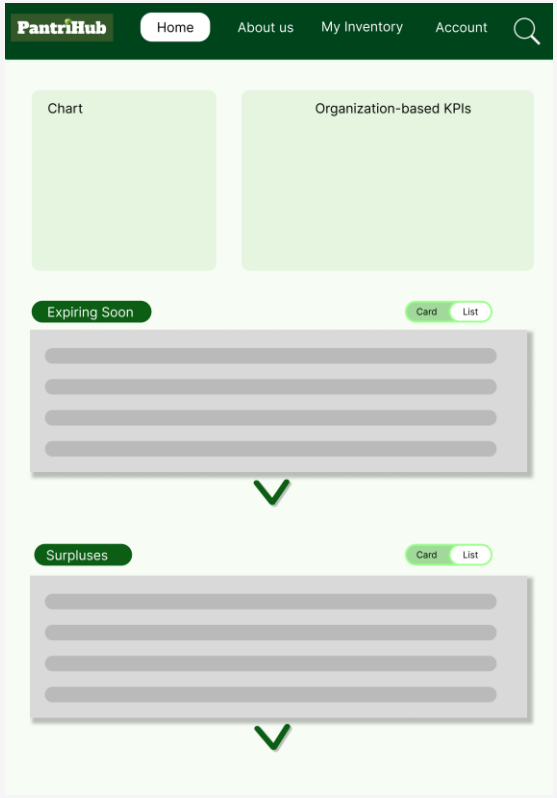

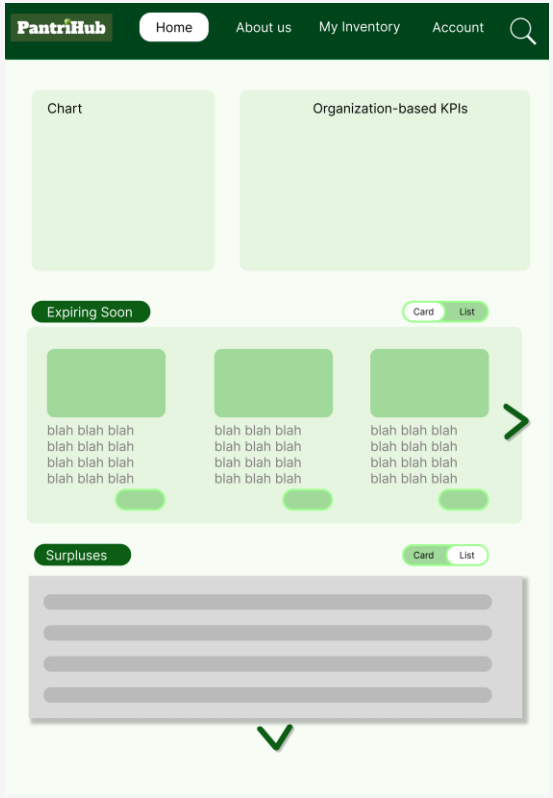

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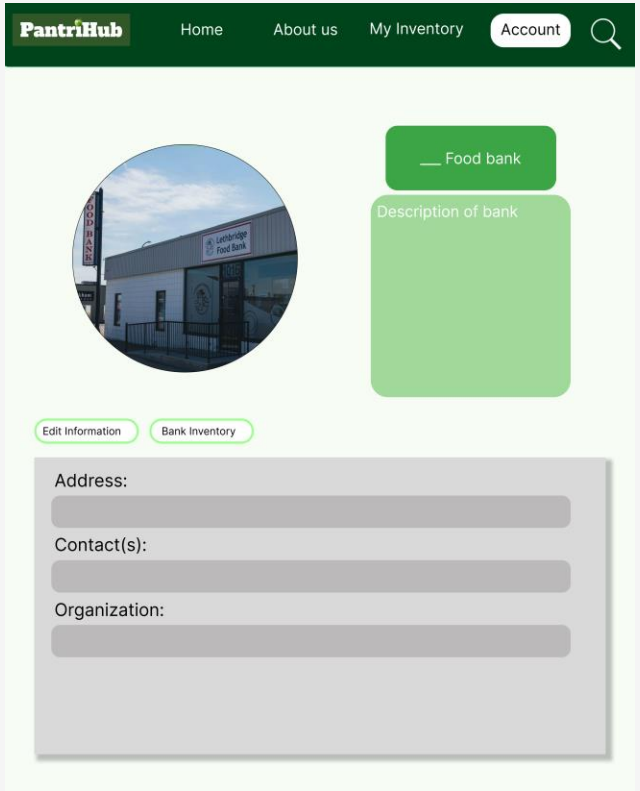

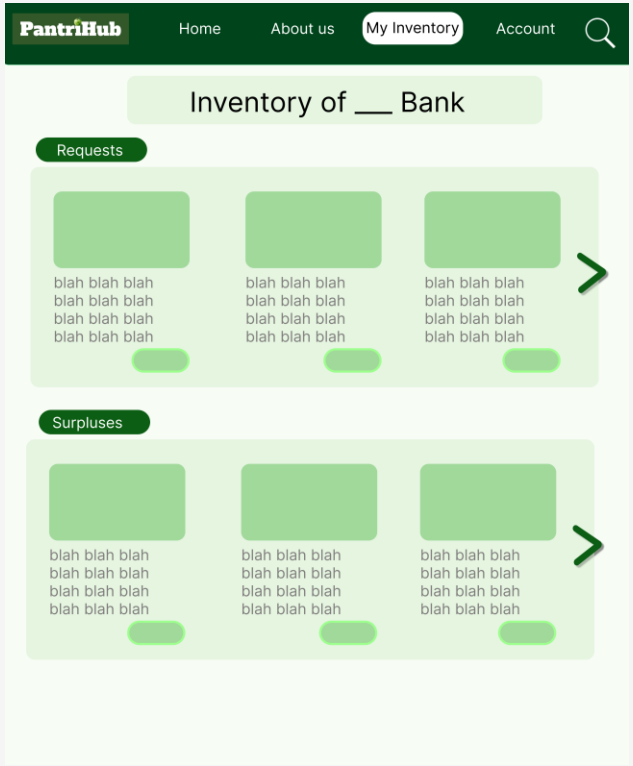

We Need Your Feedback!

We've set up a Google Survey with hi-fi designs. The survey walks you through key user flows and asks for your honest feedback.

What we're testing:

- Registering a new food bank account
- Adding a new inventory item
- Finding items expiring within 7 days
- Searching for items at a specific location
- Viewing the shared cross-location dashboard

Does our design answer: "who has what, and where?"

Scan to open survey

Please be honest -- we'd rather hear what's broken now!

Self-Reflection

How did we feel about this activity?

Bridging academic documentation with real-world design pushed us to think more carefully about who we're building for.

What are we most proud of?

Grounding every design decision in actual food bank interviews. Our stakeholder engagement isn't theoretical - it's validated.

What did we learn about collaboration?

Dividing documentation while keeping design coherent requires strong communication. The RACI matrix helped clarify ownership.

How will we apply this going forward?

User-first thinking and structured project management will carry into our development sprints.

Where do we want more help?

Feedback on our hi-fi designs, especially the shared dashboard UX and how to best surface expiry alerts.

Questions & Feedback

Group B - PantriHub

We'd love your feedback on our design direction -- scan the QR code to fill out our survey!

Scan for survey

github.com/bshra7/Inventory-Management-System-for-Food-Banks